BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Annual Examination 2026 — ANSWER KEY (Version 1)

Mathematics — SSC-I (9th Grade)

Syllabus: Unit 4 — Factorization & Algebraic Manipulation | Unit 5 — Linear Equations & Inequalities | Unit 7 — Coordinate Geometry

SECTION A — MCQ Answers (12 × 1 = 12 Marks)

Q1. The factorization of $x^4 + 4y^4$.

Correct: A $(x^2 + 2y^2 + 2xy)(x^2 + 2y^2 - 2xy)$ 1

Add & subtract $4x^2y^2$ to complete the square: $x^4 + 4x^2y^2 + 4y^4 - 4x^2y^2 = (x^2 + 2y^2)^2 - (2xy)^2 = (x^2 + 2y^2 + 2xy)(x^2 + 2y^2 - 2xy)$.

Q2. Factors of $y^2 - 7y + 12$.

Correct: A $(y - 3)(y - 4)$ 1

Find two numbers with product 12 and sum −7: −3 and −4. So $y^2 - 7y + 12 = (y - 3)(y - 4)$.

Q3. HCF of $18a^3b^4c^5$ and $24a^2b^5c^3$.

Correct: A $6a^2b^4c^3$ 1

HCF of 18, 24 = 6. Lowest power of a is 2, of b is 4, of c is 3. → HCF = $6a^2b^4c^3$.

Q4. The product of HCF and LCM of two polynomials.

Correct: C Their product 1

Key identity: HCF × LCM = P × Q (the product of the two polynomials). This is a direct analogue of the same rule for integers.

Q5. Square root of $9a^2 + 12ab + 4b^2$.

Correct: A $\pm(3a + 2b)$ 1

$9a^2 + 12ab + 4b^2 = (3a)^2 + 2(3a)(2b) + (2b)^2 = (3a + 2b)^2$. Square root of a perfect square $P = Q^2$ is $\pm Q$, so the root is $\pm(3a + 2b)$.

Q6. Factorization of $a^3 + 27b^3$.

Correct: A $(a + 3b)(a^2 - 3ab + 9b^2)$ 1

Using sum of cubes: $a^3 + (3b)^3 = (a + 3b)(a^2 - a(3b) + (3b)^2) = (a + 3b)(a^2 - 3ab + 9b^2)$.

Q7. Linear equation in one variable.

Correct: C $2x + 3 = 0$ 1

A linear equation in one variable has the standard form $ax + b = 0$ with $a \neq 0$. The exponent of x must be 1 and no products of variables. Option A has x^2 (not linear), B has xy (product), D has x^2.

Q8. Solution set of $|x + 1| = 2$.

Correct: A $\{1, -3\}$ 1

$|x + 1| = 2$ gives two cases: $x + 1 = 2 \rightarrow x = 1$, and $x + 1 = -2 \rightarrow x = -3$. Solution set = $\{1, -3\}$.

Q9. Solution set of $\sqrt{(5x)} = -10$.

Correct: B { } (empty set, φ) 1

Key fact: a square root (radical) is always non-negative for real numbers. Since the right side is −10 (negative), the equation has no real solution. Solution set = φ.

Q10. Distance between A(−1, 3) and B(4, 15).

Correct: B 13 1

$d = \sqrt{(4 - (-1))^2 + (15 - 3)^2} = \sqrt{25 + 144} = \sqrt{169} = 13$.

Q11. Minimum number of sides in a polygon.

Correct: B   3 1

A polygon is a closed figure made of line segments. A triangle (3 sides) is the polygon with the fewest sides possible.

Q12. Midpoint of (2, 4) and (6, 10).

Correct: C   (4, 7) 1

By the midpoint formula: $((2+6)/2, (4+10)/2) = (4, 7)$.

SECTION B — Short Answers (11 × 3 = 33 Marks)

Q 2(i)

Factorize: (a) $x^2 - 8x + 15$ (b) $3x^2 + 7x + 2$ 3

(a) Find two numbers with product 15 and sum −8: −3 and −5.

$x^2 - 8x + 15 = (x - 3)(x - 5)$ **1.5**

(b) Using ac method: $a \cdot c = 3 \times 2 = 6$. Numbers with product 6 and sum 7: 6 and 1.

$3x^2 + 7x + 2 = 3x^2 + 6x + x + 2 = 3x(x + 2) + 1(x + 2) = (x + 2)(3x + 1)$ **1.5**

OR

Factorize: (a) $x^2 - 4x - 5$ (b) $2x^2 + 9x + 4$ 3

(a) Numbers with product −5 and sum −4: −5 and 1.

$x^2 - 4x - 5 = (x - 5)(x + 1)$ **1.5**

(b) $a \cdot c = 2 \times 4 = 8$. Numbers with product 8, sum 9: 8 and 1.

$2x^2 + 9x + 4 = 2x^2 + 8x + x + 4 = 2x(x + 4) + 1(x + 4) = (x + 4)(2x + 1)$ **1.5**

Q 2(ii)

Factorize: $a^3 + 27$ 3

Recognize sum of cubes: $a^3 + 3^3$.

Apply $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$: **1**

$a^3 + 27 = a^3 + 3^3 = (a + 3)(a^2 - 3a + 9)$ **2**

OR

Factorize: $8x^3 - 125y^3$ 3

Recognize difference of cubes: $(2x)^3 - (5y)^3$.

Apply $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$: **1**

$8x^3 - 125y^3 = (2x - 5y)((2x)^2 + (2x)(5y) + (5y)^2) = (2x - 5y)(4x^2 + 10xy + 25y^2)$ **2**

Q 2(iii)

Find the HCF of $12x^2y^3$ and $18xy^2$ by factorization. 3

HCF of coefficients 12 and 18: HCF(12, 18) = 6. **1**

Lowest power of x: min(2, 1) = 1 → x.

Lowest power of y: min(3, 2) = 2 → y^2.

HCF = $6xy^2$ **2**

OR

Find the HCF of $a^2 - 4$ and $a^2 + 4a + 4$ by factorization. 3

Factorize each:

$a^2 - 4 = (a - 2)(a + 2)$

$a^2 + 4a + 4 = (a + 2)^2 = (a + 2)(a + 2)$ **2**

Common factor with lowest power: $(a + 2)^1$.

HCF = $a + 2$ **1**

Q 2(iv)

Find the LCM of $x^2 - 1$ and $x^2 + 2x + 1$. 3

Factorize each:

$x^2 - 1 = (x - 1)(x + 1)$

$x^2 + 2x + 1 = (x + 1)^2 = (x + 1)(x + 1)$ 1.5

LCM = product of all factors with highest power.

LCM = $(x - 1)(x + 1)^2$ 1.5

OR

Find the LCM of $6a^2b$ and $8ab^3$. 3

LCM of coefficients 6 and 8: LCM(6, 8) = 24. 1

Highest power of a: max(2, 1) = 2 → a^2.

Highest power of b: max(1, 3) = 3 → b^3.

LCM = $24a^2b^3$ 2

Q 2(v)

Find the square root of $4x^2 + 12xy + 9y^2$ by factorization. 3

Express as a perfect square:

$4x^2 + 12xy + 9y^2 = (2x)^2 + 2(2x)(3y) + (3y)^2 = (2x + 3y)^2$ 2

$\sqrt{4x^2 + 12xy + 9y^2} = \pm(2x + 3y)$ 1

OR

Simplify: $(x^2-4)/(x^2-5x+6) \times (x-3)/(x+2)$ 3

Factor each expression:

$x^2 - 4 = (x - 2)(x + 2)$

$x^2 - 5x + 6 = (x - 2)(x - 3)$ 1.5

Substitute and cancel:

$= [(x - 2)(x + 2) / ((x - 2)(x - 3))] \times [(x - 3)/(x + 2)]$

$= ((x + 2)(x - 3)) / ((x - 3)(x + 2)) = \mathbf{1}$ 1.5

Q 2(vi)

Solve for x: $3x/5 - 1/2 = x/4 + 1$ 3

LCM of 5, 2, 4 = 20. Multiply throughout by 20:

$20(3x/5) - 20(1/2) = 20(x/4) + 20(1)$

$12x - 10 = 5x + 20$ 1.5

$12x - 5x = 20 + 10 \rightarrow 7x = 30$

$x = 30/7$ 1.5

OR

Solve for x: $5x - 2 - x = 4 - 3x - 27$ 3

Simplify both sides:

$4x - 2 = -3x - 23$ 1

Collect x terms on left, constants on right:

$4x + 3x = -23 + 2 \rightarrow 7x = -21$

$x = -3$ 2

Q 2(vii)

Solve the absolute value equation: $|2x + 5| = 11$ **3**

Remove absolute value bars and write two equations:

$2x + 5 = 11$ or $2x + 5 = -11$ **1**

Solve both:

$2x = 6 \rightarrow x = 3$ | $2x = -16 \rightarrow x = -8$ **1.5**

Solution set = $\{3, -8\}$ **0.5**

OR

Solve the absolute value equation: $|3x + 2| = 7$ **3**

Write two equations:

$3x + 2 = 7$ or $3x + 2 = -7$ **1**

Solve both:

$3x = 5 \rightarrow x = 5/3$ | $3x = -9 \rightarrow x = -3$ **1.5**

Solution set = $\{5/3, -3\}$ **0.5**

Q 2(viii)

Solve: $3(-4 + 5y) \leq -8(1 - 2y) + 6$ **3**

Expand both sides:

$-12 + 15y \leq -8 + 16y + 6$

$-12 + 15y \leq -2 + 16y$ **1.5**

Collect y terms: $15y - 16y \leq -2 + 12$

$-y \leq 10 \rightarrow y \geq -10$ (inequality flips when multiplied by -1) **1**

Solution set = $\{y \mid y \in \mathbb{R} \wedge y \geq -10\}$ **0.5**

OR

Solve: $8x + 9 < 6x - 7$ **3**

Collect x terms on one side:

$8x - 6x < -7 - 9$ **1**

$2x < -16 \rightarrow x < -8$ **1.5**

Solution set = $\{x \mid x \in \mathbb{R} \wedge x < -8\}$ **0.5**

Q 2(ix)

Find the distance between $(-2, 1)$ and $(1, 5)$. **3**

By distance formula: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

$d = \sqrt{(1 - (-2))^2 + (5 - 1)^2} = \sqrt{3^2 + 4^2}$ **1.5**

$= \sqrt{9 + 16} = \sqrt{25} =$ **5 units** **1.5**

OR

Find the distance between $(0, -5)$ and $(-2, 1)$. **3**

$d = \sqrt{(-2 - 0)^2 + (1 - (-5))^2} = \sqrt{(-2)^2 + 6^2}$ **1.5**

$= \sqrt{4 + 36} = \sqrt{40} =$ **$2\sqrt{10}$ units** **1.5**

Q 2(x)

Find the midpoint of the segment joining (2, 5) and (6, 9). 3

By midpoint formula: $M = ((x_1 + x_2)/2, (y_1 + y_2)/2)$ 1

$M = ((2 + 6)/2, (5 + 9)/2) = (8/2, 14/2)$ 1

$= \mathbf{(4, 7)}$ 1

OR

Find the midpoint of the segment joining (−1, 0) and (2, 2). 3

$M = ((x_1 + x_2)/2, (y_1 + y_2)/2)$

$M = ((-1 + 2)/2, (0 + 2)/2) = (1/2, 2/2)$ 2

$= \mathbf{(1/2, 1)}$ 1

Q 2(xi)

Show that A(3, 2), B(4, 4), C(5, 6) are collinear. 3

Compute distances:

$AB = \sqrt{(4 - 3)^2 + (4 - 2)^2} = \sqrt{1 + 4} = \sqrt{5}$

$BC = \sqrt{(5 - 4)^2 + (6 - 4)^2} = \sqrt{1 + 4} = \sqrt{5}$

$AC = \sqrt{(5 - 3)^2 + (6 - 2)^2} = \sqrt{4 + 16} = \sqrt{20} = 2\sqrt{5}$ 2

Check: $AB + BC = \sqrt{5} + \sqrt{5} = 2\sqrt{5} = AC$ √

Since AB + BC = AC, points A, B, C are **collinear**. 1

OR

If (1, 4) is the midpoint of the segment joining (6, b) and (a, 2), find a and b. 3

By midpoint formula: $((6 + a)/2, (b + 2)/2) = (1, 4)$ 1

Equate coordinates:

$(6 + a)/2 = 1 \rightarrow 6 + a = 2 \rightarrow \mathbf{a = -4}$

$(b + 2)/2 = 4 \rightarrow b + 2 = 8 \rightarrow \mathbf{b = 6}$ 2

SECTION C — Long Answers (4 × 5 = 20 Marks)

Q 3(i)

Factorize: $(x + 5)(x + 3)(x + 2)(x + 6) - 88$ 5

Pair factors so that sums are equal: 5 + 3 = 8 and 2 + 6 = 8.

$= [(x + 5)(x + 3)]\,[(x + 2)(x + 6)] - 88$

$= (x^2 + 8x + 15)(x^2 + 8x + 12) - 88$ 1.5

Let $x^2 + 8x = a$. Expression becomes:

$(a + 15)(a + 12) - 88 = a^2 + 27a + 180 - 88 = a^2 + 27a + 92$ 1.5

Factor $a^2 + 27a + 92$: numbers with product 92, sum 27 → 23 and 4.

$= a^2 + 4a + 23a + 92 = a(a + 4) + 23(a + 4) = (a + 4)(a + 23)$ 1

Substitute back $a = x^2 + 8x$:

$= \mathbf{(x^2 + 8x + 4)(x^2 + 8x + 23)}$ 1

OR

Factorize: $(a^2 - 5)^2 - 13(a^2 - 5) + 36$ 5

Let $a^2 - 5 = y$. Expression becomes:

$y^2 - 13y + 36$ 1

Numbers with product 36, sum −13 → −4 and −9.

$y^2 - 13y + 36 = (y - 4)(y - 9)$ 2

Substitute $y = a^2 - 5$:

$= (a^2 - 5 - 4)(a^2 - 5 - 9) = (a^2 - 9)(a^2 - 14)$

Factor further $a^2 - 9 = (a - 3)(a + 3)$:

$= \mathbf{(a - 3)(a + 3)(a^2 - 14)}$ 2

Q 3(ii)

Find the square root of $x^4 + 8x^3 + 20x^2 + 16x + 4$ by division method. [5]

Arrange in descending order (already done). Start the long division pattern:

```
                 x² + 4x + 2
              ┌──────────────────────────
        x²    │ x⁴ + 8x³ + 20x² + 16x + 4
              │ x⁴
         ─────┼─────
      2x²+4x  │      8x³ + 20x²
              │      8x³ + 16x²
         ─────┼─────
   2x²+8x+2   │             4x² + 16x + 4
              │             4x² + 16x + 4
         ─────┼─────
                                        0
```

Successive partial roots build up: $x^2 \rightarrow x^2 + 4x \rightarrow x^2 + 4x + 2$. [3]

Remainder is 0, so the expression is a perfect square.

Square root = $\pm(x^2 + 4x + 2)$ [2]

OR

Find the HCF of $12 + 16a + 7a^2 + a^3$ and $13a + 5a^2 + 14 + a^3$ by division method. [5]

Arrange both in descending order:

$P = a^3 + 7a^2 + 16a + 12$; $Q = a^3 + 5a^2 + 13a + 14$ [1]

Divide P by Q (since same degree, either can be divisor):

$P - Q = (a^3 + 7a^2 + 16a + 12) - (a^3 + 5a^2 + 13a + 14) = 2a^2 + 3a - 2$ [1.5]

Now divide Q by the new remainder $2a^2 + 3a - 2$ (multiplying for convenience if needed).

After successive divisions, we obtain the last non-zero remainder:

HCF = $a + 2$ [2.5]

Q 3(iii)

Solve the compound inequality: $2x + 3 > 7$ OR $4x - 1 < 3$ ($x \in \mathbb{R}$). [5]

Solve each inequality separately:

$2x + 3 > 7 \rightarrow 2x > 4 \rightarrow x > 2$ [1.5]

$4x - 1 < 3 \rightarrow 4x < 4 \rightarrow x < 1$ [1.5]

For compound "OR", take the **union** of solution sets.

Solution set = $\{x \mid x \in \mathbb{R} \wedge (x > 2 \text{ OR } x < 1)\}$ [1]

On a number line: all real numbers less than 1, or greater than 2 (open circles at 1 and 2). [1]

OR

Solve: $x - 5 \geq -1$ AND $x + 3 \leq 10$ ($x \in \mathbb{R}$). [5]

Solve each inequality:

$x - 5 \geq -1 \rightarrow x \geq 4$ [1.5]

$x + 3 \leq 10 \rightarrow x \leq 7$ [1.5]

For compound "AND", take the **intersection**.

Solution set = $\{x \mid x \in \mathbb{R} \wedge 4 \leq x \leq 7\}$ [1]

On a number line: closed circles at 4 and 7 with the segment between them shaded. [1]

Q 3(iv)

Show that A(−4, 2), B(1, 4), C(3, −1), D(−2, −3) are the vertices of a square. 5

A square has (i) all four sides equal and (ii) equal diagonals.

Compute side lengths:

$AB = \sqrt{(1+4)^2 + (4-2)^2} = \sqrt{25 + 4} = \sqrt{29}$

$BC = \sqrt{(3-1)^2 + (-1-4)^2} = \sqrt{4 + 25} = \sqrt{29}$

$CD = \sqrt{(-2-3)^2 + (-3+1)^2} = \sqrt{25 + 4} = \sqrt{29}$

$DA = \sqrt{(-4+2)^2 + (2+3)^2} = \sqrt{4 + 25} = \sqrt{29}$ 3

All four sides equal: $AB = BC = CD = DA = \sqrt{29}$ ✓

Compute diagonals:

$AC = \sqrt{(3+4)^2 + (-1-2)^2} = \sqrt{49 + 9} = \sqrt{58}$

$BD = \sqrt{(-2-1)^2 + (-3-4)^2} = \sqrt{9 + 49} = \sqrt{58}$ 1.5

Both diagonals equal: $AC = BD = \sqrt{58}$ ✓

Since all sides are equal AND diagonals are equal, ABCD is a **square**. 0.5

OR

Show that A(−4, −1), B(0, −2), C(6, 1), D(2, 2) are the vertices of a parallelogram. 5

A parallelogram has both pairs of opposite sides equal (or equivalently, diagonals bisect each other).

Compute side lengths:

$AB = \sqrt{(0+4)^2 + (-2+1)^2} = \sqrt{16 + 1} = \sqrt{17}$

$BC = \sqrt{(6-0)^2 + (1+2)^2} = \sqrt{36 + 9} = \sqrt{45}$

$CD = \sqrt{(2-6)^2 + (2-1)^2} = \sqrt{16 + 1} = \sqrt{17}$

$DA = \sqrt{(-4-2)^2 + (-1-2)^2} = \sqrt{36 + 9} = \sqrt{45}$ 3

Opposite sides: $AB = CD = \sqrt{17}$ ✓ and $BC = DA = \sqrt{45}$ ✓ 1

Check: midpoint of AC = $((-4+6)/2, (-1+1)/2) = (1, 0)$.

Midpoint of BD = $((0+2)/2, (-2+2)/2) = (1, 0)$. Diagonals bisect each other ✓

Since opposite sides are equal and diagonals bisect, ABCD is a **parallelogram**. 1